

Name _____ Date _____ Period _____

When Is an Inverse a Function?

Algebra II Honors | Unit 1 | Day 12 | TEK A2.2.C

Objective: I can determine whether the inverse of a function is a function, choose a domain restriction that makes it one, and apply this to a quadratic whose vertex is not at the origin.

Vocabulary

One-to-one — every output comes from exactly one input. Detected by the _____.

Key fact — the inverse of a function is always a _____, but a _____ only when the original is one-to-one.

Domain restriction — deliberately shrinking the domain to force one-to-one behavior. Restricting the domain of f restricts the _____ of f^{-1} .

Part 1 — Why $f(x) = x^2$ Fails

x	-3	-2	-1	0	1	2	3
$f(x) = x^2$							

Repeated outputs: _____ So f is _____

Swapping produces both $(1, 1)$ and $(1, -1)$. State precisely which test the inverse fails and why:

The _____ — the input 1 would need two outputs.

Solving $y = x^2$ for x gives $x = \pm \sqrt{y}$, and the _____ is the whole problem.

Part 2 — Restrict and Repair

Restriction on x^2	Half of the parabola kept	Resulting $f^{-1}(x)$	Range of f^{-1}
$x \geq 0$	_____	_____	_____
$x \leq 0$	_____	_____	_____

Both restrictions work. Is one more correct than the other? _____

Part 3 — A Vertex That Is Not at the Origin

Now take $g(x) = (x - 2)^2$. The same idea applies, but you must locate the turning point first.

Vertex of g : _____ So the natural restriction is _____

Now find the inverse. Swap and solve for y , keeping track of the $\pm$.

$x = (y - 2)^2 \rightarrow \pm \sqrt{y - 2} = y - 2 \rightarrow y = \pm \sqrt{y - 2} + 2$

With the restriction $x \geq 2$ on g , the inverse is $g^{-1}(x) = \sqrt{x - 2} + 2$ Domain of g^{-1} _____

Why does the 2 appear OUTSIDE the radical in the inverse when it was inside the square in g?

Undoing happens in _____: take the root first, then add 2.

Part 4 — Which Need No Restriction?

Function	One-to-one?	Restriction needed?	Inverse
x^2	_____	_____	_____
x^3	_____	_____	_____
2^x	_____	_____	_____
$ x $	_____	_____	_____
$1/x$	_____	_____	_____

State the pattern as a rule about increasing and decreasing:

A function that only increases or only decreases is _____; one that turns around is not.

Part 5 — Going Further

1. Restrict $h(x) = (x + 3)^2$ so its inverse is a function, then write the inverse.

Restrict _____ $h^{-1}(x) =$ _____

2. Explain why $1/x$ is its own inverse, using the swap.

3. Is there a restriction of $|x|$ whose inverse is a function? Give one. _____

Exit Ticket

Consider $f(x) = (x - 5)^2$ with no restriction.

Is its inverse a function? _____ Give a restriction that fixes it: _____

With that restriction, $f^{-1}(x) =$ _____ Explain in one sentence why the restriction is needed:
